
USB-6501 User Manual

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Using the Digital Input/Output Lines

The USB-6501 device has 24 single-ended digital lines.

The USB-6501 features three DIO ports:

- P0.<0..7>
- P1.<0..7>
- P2.<0..7>

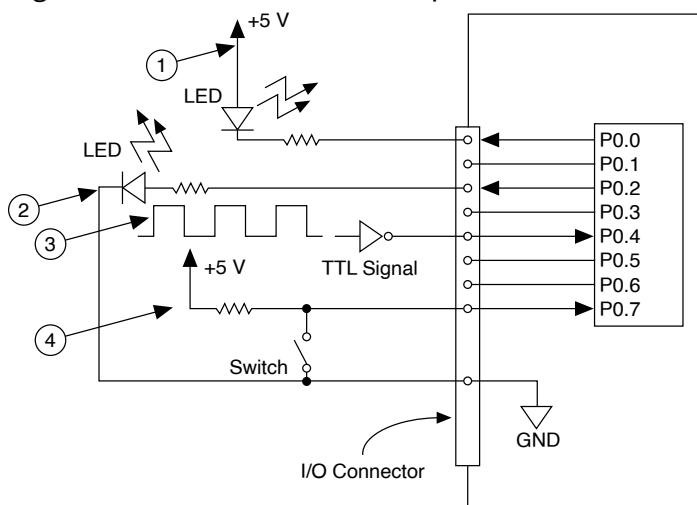


Note P2.7/PFI 0 can also function as a 32-bit event counter.

Digital I/O Circuitry

The following figure shows some common examples of connections of DIO lines with standard circuits. In the figure, P0.<0..7> is connected to example signals configured as digital inputs and digital outputs. You can configure P1.<0..7> and P2.<0..7> similarly.

Figure 8. DIO Connections Example



1. P0.0 configured as an open collector (open-drain) digital output driving an LED
2. P0.2 configured as an active drive (push-pull) digital output driving an LED
3. P0.4 configured as a digital input receiving a TTL signal from a gated inverter
4. P0.7 configured as a digital input receiving a 0 V or 5 V signal from a switch



Caution Exceeding the maximum input voltage ratings or maximum output ratings listed in the device specifications can damage the DAQ device and the computer. National Instruments is not liable for any damage resulting from such signal connections.

Related information:

- [USB-6501 Specifications](#)